Comparison of No-Reference Image Quality Models via MAP Estimation in Diffusion Latents

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Motivation

- analysis-by-synthesis framework
- NR-IQA的感知优化视角: NR-IQA能否用于图像的优化？
  - $q_{\boldsymbol{w}}(\cdot) : \mathbb{R}^N \mapsto \mathbb{R},$ an NR-IQA model

$$\boldsymbol{x}^\star = \arg\max_{\boldsymbol{x}} q_{\boldsymbol{w}}(\boldsymbol{x}),$$

  - 如果直接优化，现有的NR-IQA表现不佳

(a) $q_{\boldsymbol{w}}(\boldsymbol{x}) = 2.49$ **(b)** $q_{\boldsymbol{w}}(\boldsymbol{x}^\star) = 5.00$

Method

- 最大后验（MAP）视角

$$p(\boldsymbol{x}|\boldsymbol{x}^{\text{init}}) \propto \exp(-E(\boldsymbol{x}|\boldsymbol{x}^{\text{init}}))$$

$$\boldsymbol{x}^\star = \arg\min_{\boldsymbol{x}} E(\boldsymbol{x}|\boldsymbol{x}^{\text{init}}) = \arg\min_{\boldsymbol{x}} D(\boldsymbol{x}, \boldsymbol{x}^{\text{init}}) - \lambda q_{\boldsymbol{w}}(\boldsymbol{x})$$

  - $D(\cdot,\cdot) : \mathbb{R}^{N\times N} \mapsto \mathbb{R}$ fidelity term

(a) $q_{\boldsymbol{w}}(\boldsymbol{x}) = 2.49$ **(b)** $q_{\boldsymbol{w}}(\boldsymbol{x}^\star) = 5.00$ **(c)** $q_{\boldsymbol{w}}(\boldsymbol{x}^\star) = 5.00$ **(d)** $q_{\boldsymbol{w}}(\boldsymbol{x}^\star) = 5.00$

Method

- 如何更好地估计最大后验？
  - 图像增强中的传统做法

$$\begin{aligned}\boldsymbol{x}^{\star} &= \arg\max_{\boldsymbol{x}} p(\boldsymbol{x}|\boldsymbol{x}^{\text{init}}) = \arg\max_{\boldsymbol{x}} p(\boldsymbol{x}^{\text{init}}|\boldsymbol{x})p(\boldsymbol{x}) \\ &= \arg\min_{\boldsymbol{x}} -\underline{\log p(\boldsymbol{x}^{\text{init}}|\boldsymbol{x})} - \underline{\log p(\boldsymbol{x})}.\end{aligned}$$

图像的自然度
（包含图像的先验信息）

相对于初始图像的保真度
（MSE，FRIQA）

Method

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图像的自然度
（包含图像的先验信息）

图像的保真度
（MSE，FRIQA）

  - 最大后验中，如何利用NR-IQA中的先验信息？

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NR-IQA

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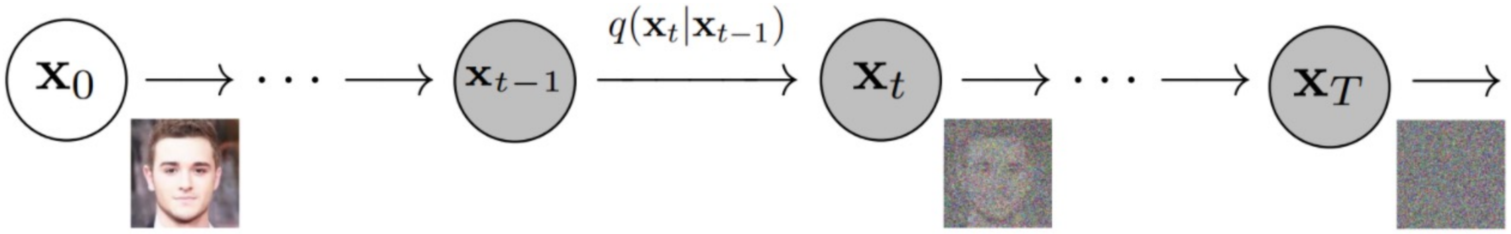
图像的保真度
（MSE，FRIQA）

  - 最大后验中，如何利用NR-IQA中的先验信息？

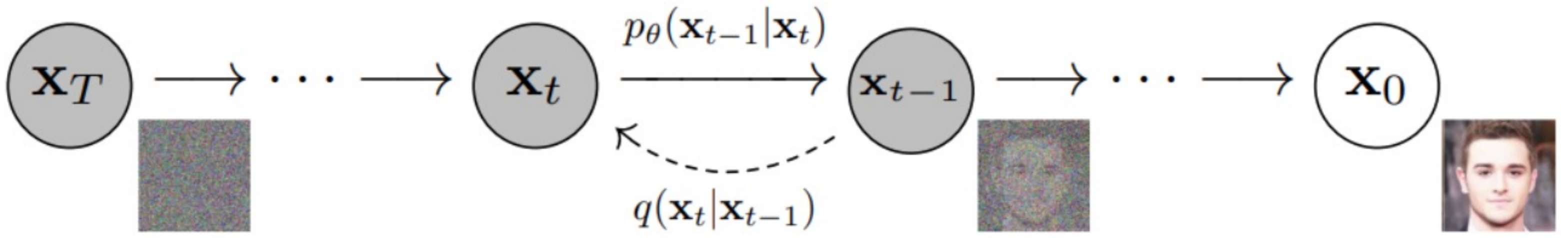
latents in diffusion model

NR-IQA

- 为图像逐步加入高斯噪声，这一过程符合Markov过程：



- Diffusion model-从噪声建立自然图像的过程



Method

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diffusion model $\boldsymbol{h}(\cdot)$

NR-IQA

从噪声恢复图像的过程

$$\boldsymbol{x}_T^{\star} = \arg\min_{\boldsymbol{x}_T} D\left(\boldsymbol{h}(\boldsymbol{x}_T), \boldsymbol{x}^{\text{init}}\right) - \lambda q_{\boldsymbol{w}}(\boldsymbol{h}(\boldsymbol{x}_T))$$

$D(\cdot,\cdot)$：取diffusion latents中的MSE

Method

$$\boldsymbol{x}_T^\star = \arg\min_{\boldsymbol{x}_T} D\left(\boldsymbol{h}(\boldsymbol{x}_T), \boldsymbol{x}^{\text{init}}\right) - \lambda q_{\boldsymbol{w}}(\boldsymbol{h}(\boldsymbol{x}_T))$$

- 保真度（$D(\cdot,\cdot)$）的重要性

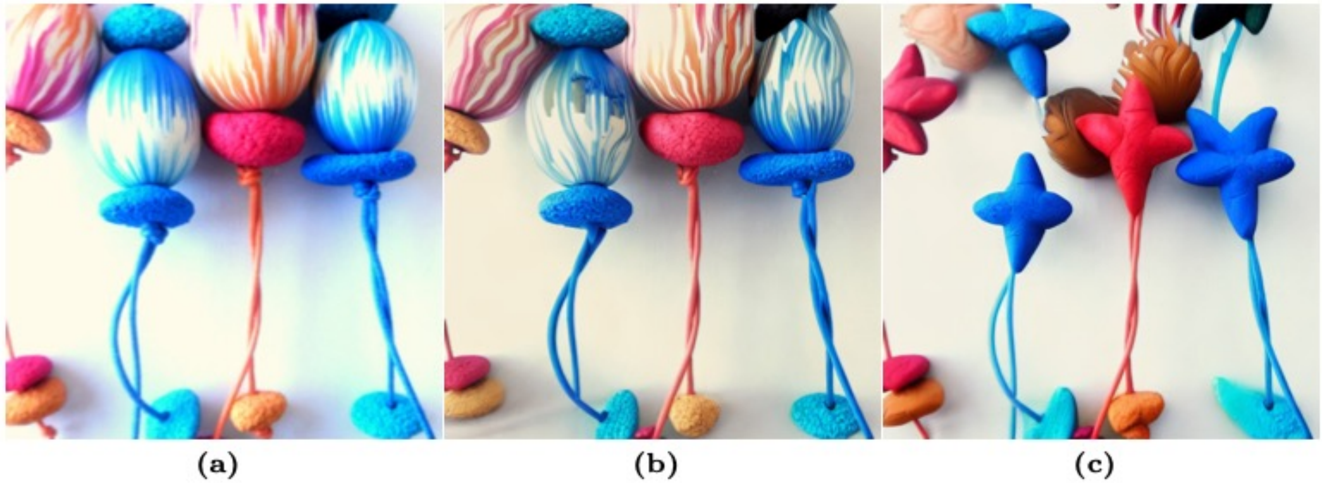

Fig. 2: (a) Initial image with slight over-exposure degradation. **(b)** MAP-enhanced image of (a) by Algorithm 1, in which the NR-IQA model LIQE [85] is adopted with $\lambda = 0.1$. **(c)** MAP-enhanced image with LIQE and $\lambda = 1.0$, whose semantics undesirably deviate from (a).

Experiment

- Diffusion model： EDICT
- 扩散步数 T=50

- 20张真实失真图像
- 8种NR-IQA方法
- 25个被试对生成的图像进行评价

Experiment

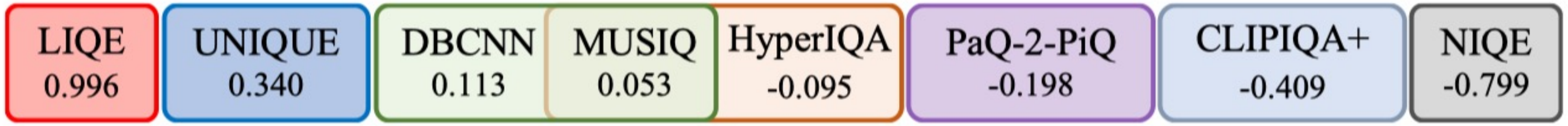

Fig. 5: Subjective ranking of NR-IQA models using MAP estimation in diffusion latents. Below each model is the averaged quality score over 20 test image (larger is better). Models within the same colored box have statistically indistinguishable performance.

NR-IQA Model	Training Set	# of Params	SRCC Rank	MAP Rank	$\triangle$ Rank
NIQE [42]	–	0.001	8 (0.706)	8	0
CLIPIQA+ [68]	KonIQ-10k	101.349	2 (0.855)	7	-5
PaQ-2-PiQ [78]	FLIVE	11.704	5 (0.827)	6	-1
HyperIQA [60]	KonIQ-10k	27.375	7 (0.776)	5	2
MUSIQ [30]	KonIQ-10k	27.125	3 (0.853)	4	-1
DBCNN [82]	KonIQ-10k	15.311	6 (0.789)	3	3
UNIQUE [83]	Multiple	22.322	4 (0.838)	2	2
LIQE [85]	Multiple	150.976	1 (0.881)	1	0

Experiment

(a) Initial (b) NIQE [42] (c) DBCNN [82]

(d) HyperIQA [60] (e) PaQ-2-PiQ [78] (f) UNIQUE [83]

(g) MUSIQ [30] (h) CLIPIQA+ [68] (i) LIQE [85]

Experiment

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Experiment

(a) LIQE (b) UNIQUE (c) DBCNN (d) MUSIQ

(e) HyperIQA (f) PaQ-2-PiQ (g) CLIPIQA+ (h) NIQE

Fig. 6: Failure cases corresponding to subcaptioned NR-IQA models. **(a)** Incomplete object. **(b)** False details. **(c)** Unnatural/cartoon texture. **(d)** Over-smoothness. **(e)** Detail drift. **(f)** Over-saturated color. **(g)** Failed enhancement. **(h)** Grainy appearance. Local artifacts are highlighted in red blocks, and are better contrasted to the initials in Fig. 4.

Writing

Contemporary no-reference image quality assessment (NRIQA) models can effectively quantify the perceived image quality, with high correlations between model predictions and human perceptual scores on fixed test sets. However, little progress has been made in comparing NR-IQA models from a perceptual optimization perspective. Here, for the first time, we demonstrate that NR-IQA models can be plugged into the maximum a posteriori (MAP) estimation framework for image enhancement. This is achieved by taking the gradients in differentiable and bijective diffusion latents rather than in the raw pixel domain. Different NR-IQA models are likely to induce different enhanced images, which are ultimately subject to psychophysical testing. This leads to a new computational method for comparing NR-IQA models within the analysis-by-synthesis framework. Compared to conventional correlationbased metrics, our method provides complementary insights into the relative strengths and weaknesses of the competing NR-IQA models in the context of perceptual optimization.

1️⃣NR-IQA已有较大进展

2️⃣目前的问题

Writing

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3️⃣本文用什么方法，提出了解决方案

Writing-Abstract

Contemporary no-reference image quality assessment (NRIQA) models can effectively quantify the perceived image quality, with high correlations between model predictions and human perceptual scores on fixed test sets. However, little progress has been made in comparing NR-IQA models from a perceptual optimization perspective. Here, for the first time, we demonstrate that NR-IQA models can be plugged into the maximum a posteriori (MAP) estimation framework for image enhancement. This is achieved by taking the gradients in differentiable and bijective diffusion latents rather than in the raw pixel domain. Different NR-IQA models are likely to induce different enhanced images, which are ultimately subject to psychophysical testing. This leads to a new computational method for comparing NR-IQA models within the analysis-by-synthesis framework. Compared to conventional correlationbased metrics, our method provides complementary insights into the relative strengths and weaknesses of the competing NR-IQA models in the context of perceptual optimization.

4️⃣具体怎么做的

5️⃣ 6️⃣工作的意义

Writing-Introduction

Image quality assessment (IQA) models have been extensively studied [70] as surrogates of human subjects for performance evaluation of image processing and computer vision systems. Depending on the accessibility to pristine-quality reference images, IQA models can be broadly categorized into two types: fullreference IQA (FR-IQA) [71,79] and no-reference IQA (NR-IQA) [2,41]. FR-IQA models are well-suited for scenarios where undistorted references are available. However, in many real-world applications (e.g., low-light image enhancement and image-to-image translation), specifying desired reference outputs is an inherently challenging task [6], if not impossible. This highlights the necessity for accurate NR-IQA models to assess the perceived image quality without referencing to the pristine-quality counterparts.

1️⃣IQA介绍和已有问题

Writing-Introduction

Apart from performance evaluation, IQA models hold great promise for perceptual optimization of image processing systems. FR-IQA models have been serving the purpose for a long time, from the age of measuring error visibility and structural similarity [71] to the data-driven era [13,79]. Recently, an extensive comparison of FR-IQA models for optimization of various image processing algorithms has been conducted [12], turning the perceptual optimization process into an analysis-by-synthesis tool [23]. A natural question then arises:

Can we compare NR-IQA models in terms of their performance in perceptual optimization as well?

A straightforward computational effort to answer the above question is directly maximizing an NR-IQA model, q w (·) : R N → ↦ R, parameterized by a vector w, with respect to the input image x:

$$\boldsymbol{x}^\star = \arg\max_{\boldsymbol{x}} q_{\boldsymbol{w}}(\boldsymbol{x}),$$

where a larger q w (·) indicates higher predicted quality, and x ⋆ ∈ R N is the optimized image. As observed in [81] and shown in Fig. 1 (b), even if q w is instantiated by a state-of-the-art NR-IQA model (LIQE [85] in this case), the optimized image appears locally texture and color distorted, with degraded quality compared to the initial.

2️⃣介绍IQA在感知优化中的应用

Writing-Introduction

A natural extension is to interpret q w (·) as a natural image prior, and plug it into the maximum a posteriori (MAP) estimation framework, leading to the following optimization problem:

$$\boldsymbol{x}^{\star} = \arg\min_{\boldsymbol{x}} E(\boldsymbol{x}|\boldsymbol{x}^{\text{init}}) = \arg\min_{\boldsymbol{x}} D(\boldsymbol{x}, \boldsymbol{x}^{\text{init}}) - \lambda q_{\boldsymbol{w}}(\boldsymbol{x}),$$

where we express the posterior probability p(x | x init) ∝ exp(−E(x | x init)) by its associated Gibbs energy. D(·, ·) : R N×N → ↦ R is the fidelity term (also known as the negative log-likelihood term), which can be implemented by a distance metric. λ is a parameter, trading off the likelihood and the prior terms. As shown in Fig. 1 (c), even when λ is optimally set through careful visual inspection, the MAP-optimized image is still no better than the initial, which manifests itself as a perceptually imperceptible adversarial example of q w (·) [81].

3️⃣对上一段的问题进行转换和求解

Writing-Introduction

Here, for the first time, we successfully demonstrate the perceptual optimization capability of contemporary NR-IQA models in the MAP estimation framework (see Fig. 1 (d)). Taking inspiration from the impressive progress of diffusion models [25, 55–57, 77] in image synthesis and beyond [11, 58], we supplement an NR-IQA method with a differentiable and bijective diffusion model. This empowers the NR-IQA model with the capability of modeling highly complex distributions of natural images by operating in the diffusion latent space 4 . Specifically, we work with the method of exact diffusion inversion via coupled transformations (EDICT) [66], that turns any pre-trained diffusion model into a bijection between the input image and the latent noise vector through affine coupling [14,15].

4️⃣介绍本文的方法

Writing-Introduction

The resulting paradigm, MAP estimation in diffusion latents, gives us an opportunity to compare NR-IQA models within the analysis-by-synthesis framework. Specifically, we plug eight NR-IQA models into this paradigm to enhance a set of photographic images with realistic camera distortions [16,19,27] . Psychophysical testing on the optimized images of identical visual content reveals the relative performance of the competing NR-IQA models in image enhancement, and provides complementary insights into future NR-IQA model development beyond fixed-set evaluation.

5️⃣介绍自己方法的应用价值

In summary, this paper presents two main contributions.

– We propose the method of MAP estimation in diffusion latents, which enables "imperfect" NR-IQA models to act as natural image priors for perceptual optimization.

– We apply the proposed method to compare eight contemporary NR-IQA models with a number of interesting observations.

6️⃣总结